

Wajeeh Alam

647-285-7970 | w5alam@uwaterloo.ca | linkedin.com/in/wajeeh-alam | github.com/wajeeh-alam

EDUCATION

University of Waterloo

Sept. 2026 – 2031

Bachelor of Computer Science (Honours Co-op)

Waterloo, ON

- **Relevant Coursework:** Designing Functional Programs (CS 135), Elementary Algorithm Design (CS 136), Algebra (MATH 135), Calculus I (MATH 137)

EXPERIENCE

Founding Engineer · Toronto, ON

Oro

Aug. 2026 – Present

- Architected and shipped an SMS-native FastAPI backend allowing **2K+ downloads** by integrating Twilio/Sendblue with PostgreSQL/pgvector & Redis for **LLM** onboarding and **9+ multimodal AI features**.
- Built a production-grade **long-term memory layer** in **PostgreSQL**, transforming preferences and user corrections into persistent conversational context
- **Reduced hallucinations by 40%** across 9+ AI workflows by enforcing schema-validated responses/fail-safe logic.

Senior Coding Instructor · Toronto, ON

University of Toronto

Jun. 2025 – Aug. 2026

- Delivered **Python** & HTML/CSS/JS instruction to **300+ students** across varied ages and skill levels by designing interactive teaching tools, workshops, and tailored curricula in game & web development projects.
- Increased student participation by **45%** and strengthened comprehension of complex programming logic by developing **interactive teaching games** with hands-on exercises and immediate feedback.

Lead Software Engineer · Waterloo, ON

SproutHacks

Jun. 2024 – Sept. 2024

- Enabled registration and event operations for **350 users** by leading development of a **React/TypeScript** portal and dashboard with Tailwind CSS and Supabase via CI/CD practices.
- Kept onboarding and dashboard data synchronized in real time for **100+ active participants** by integrating **REST APIs** and **Firebase Authentication**.

PROJECTS

drawOff | *React, TypeScript, Tailwind CSS, Node.js, Socket.io*

- Maintained **sub-second player synchronization** across concurrent rooms and round transitions by building **Socket.io** networking with server-authoritative state management.
- Unified **AI vision scoring** & spectator votes with live player votes overriding **VLM** (Gemini API) generated results if present.

Voca (MathHacks) | *React, TypeScript, Tone.js, MediaPipe*

- Built a browser-based vocal training platform that analyzed pitch in real time and converted vocal accuracy into interactive exercises and immediate feedback.
- Combined **3 real-time processing pipelines**; MediaPipe hand tracking, Tone.js audio modulation, and autocorrelation-based pitch detection; to enable hands-free vocal training.

Pokédoku | *React, TypeScript, Tailwind CSS*

- Generated valid **9×9 Sudoku boards** with 40 playable blanks per game using randomized **backtracking** and solution-backed answer validation.
- Created device-adaptive controls for desktop drag-and-drop and mobile tap-to-place, alongside a custom audio engine managing **19 music tracks** and reactive gameplay sound effects.

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, HTML/CSS, Python, Lua, C++

Frameworks/Libraries: React/React Native, Node.js, Pytest, SQL/PostgreSQL, Redis, FastAPI, Supabase

Tools/Platforms: AWS, Azure, Git/GitHub Actions, CI/CD